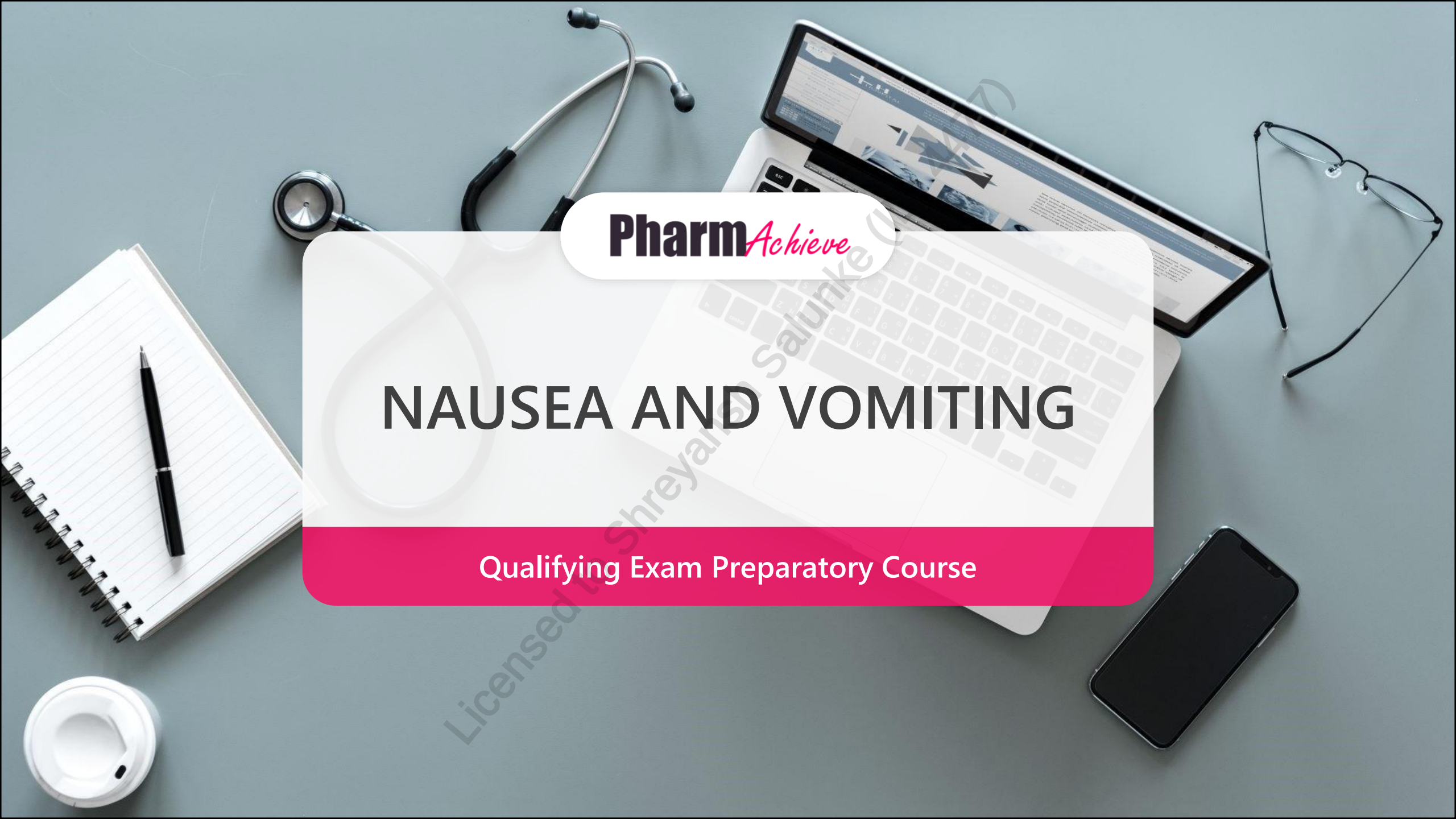




Pharm*Achieve*

NAUSEA AND VOMITING

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

BBB	Blood Brain Barrier
CINV	Chemotherapy-Induced Nausea and Vomiting
CNS	Central Nervous System
CTZ	Chemoreceptor Trigger Zone
EPS	Extrapyramidal Symptoms
HG	Hyperemesis Gravidarum
ICH	Intracranial Hemorrhage
IM	Intramuscular
IV	Intravenous
GI	Gastrointestinal
GU	Genitourinary
MAOI	Monoamine Oxidase Inhibitor
N/V	Nausea/Vomiting
NVP	Nausea and Vomiting in Pregnancy
ORS	Oral Rehydration Solution
PONV	Post-Operative Nausea and Vomiting
QID	4 times a day

OBJECTIVES

- 1 Describe the signs and symptoms of nausea and vomiting
- 2 Describe the pathophysiology and pharmacology of nausea and vomiting
- 3 Identify the possible medical and medication-related causes of nausea and vomiting
- 4 Determine the types of nausea and vomiting
- 5 Determine appropriate therapy
- 6 Discuss side effects and contraindications to specific therapy

DEFINITIONS

NAUSEA

Unpleasant sensation experienced **prior to vomiting** with various underlying causes, considered a symptom (not a diagnosis)

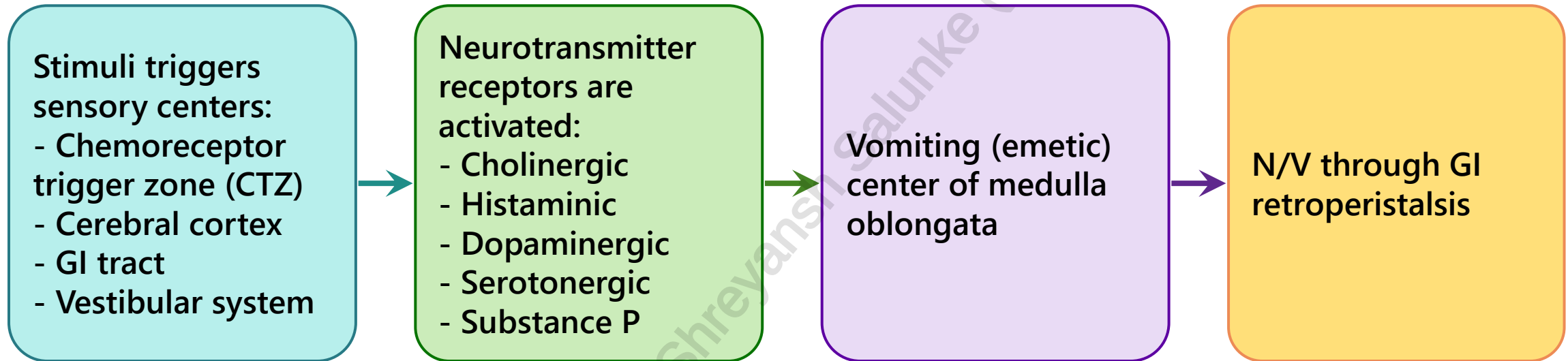
- Acute: < 1 week
- Chronic: ≥1 month

VOMITING/EMESIS

- **Oral discharge** of gastric contents

RETCHING

Muscle contraction without expulsion of gastric contents ("**gagging**" or "**dry heaving**")



CONDITION-ASSOCIATED NAUSEA AND VOMITING

GI/GU

Pancreatitis, **peptic ulcer disease**, gastric outlet/bowel obstruction, **pyelonephritis**, **gastroenteritis**, gastroparesis, **functional dyspepsia**, gastric malignancies

DRUG WITHDRAWAL

Opiates, benzodiazepines

NEUROLOGICAL

Headaches/migraines, vestibular disorders, intracranial hemorrhage (ICH), head trauma

METABOLIC

Diabetic ketoacidosis, Addison disease, renal disease

OTHERS

Hyponatremia, uremia, anxiety/depression disorders, anticipatory nausea, anorexia, **pregnancy**, otitis media, post-op, noxious odours, **acute infection**

MEDICATION-ASSOCIATED NAUSEA AND VOMITING

- Cytotoxic chemotherapy
- Radiation therapy
- Anticonvulsants
 - Carbamazepine
 - Phenytoin
 - Phenobarbital
 - Valproic acid
- Antibiotics
 - Erythromycin/clarithromycin
 - Tetracyclines
 - Anti-tuberculosis agents
- Anti-inflammatories
 - ASA
 - NSAIDs
 - Colchicine
- Cannabis
- Antidiabetics
 - Metformin
- Sulfonylurea
- Cardiovascular
 - Beta-blockers
 - Calcium channel blockers
 - Digoxin
 - Amiodarone
 - Procainamide
- Opioids
- Nicotine
- Levodopa
- Theophylline
- Oral contraceptives
- Acyclovir
- Caffeine
- Antidepressants e.g. SSRIs

* Not a comprehensive list

CLINICAL PRESENTATION

Patients may experience:

- Dehydration (*assess mucous membranes, skin turgor*)
- Abdominal pain
- Distress/Anxiety
- Discomfort
- Electrolyte abnormalities
- Taste alterations
- Weight loss

RED FLAGS – WHEN TO SEEK IMMEDIATE MEDICAL ATTENTION



- **Significant dehydration** (e.g. dizziness, sunken eyes, repeated vomiting, unable to keep liquids down for >8 hours)
- **Blood** in the vomit (hematemesis) or stool
- **Coffee ground emesis**
- Altered **mental status**, neurological deficits, recent head trauma
- **“Thunderclap”** headache (severe headache with nausea and vomiting, generally described as “the worst headache you’ve ever had”)
- **Fever and stiff neck** in addition to nausea and vomiting
- **Unintended weight loss**

SIGNS OF DEHYDRATION

CHILDREN

- Dry mouth, tongue, and skin
- Lack of tears
- Decreased urine output (<4 wet diapers in 24 hours)
- Irritability and lethargy
- Reduced skin turgor
- Sunken eyes
- Change in skin colour
- Sunken fontanel (infants)

ADULTS

- Dry mouth and tongue
- Feeling more thirsty than usual
- Weakness and lethargy
- Reduced need to urinate
- Sunken eyes

GOALS OF THERAPY

- ✓ Treat the underlying cause and/or condition
- ✓ Minimize the severity and frequency of nausea and/or vomiting
- ✓ Prevent complications and reoccurrence of N/V
- ✓ Minimize side effects of antiemetic medications
- ✓ Decrease or eliminate the day-to-day disruptions caused by nausea and/or vomiting

GENERAL MANAGEMENT

Non-Pharmacological Therapy

- Ensure **adequate hydration** (at least 2.5L of fluid daily); treat with oral rehydration solutions for moderate to severe cases of N/V or when intake is poor
- **Eat small, bland and frequent meals** and avoid stimuli (e.g. noxious odours)
- Consult dietitian when necessary (nausea associated with food intolerances)
- Avoid triggers (e.g. spicy foods, high fat foods, strong odours, etc.)

Pharmacological Therapy

- Viral gastroenteritis (food poisoning): lack of superiority of one agent over another
- Vestibular → antihistamines, anticholinergics
- Problems with GI motility → **metoclopramide or domperidone**
- CTZ stimulation → **dopamine antagonists, serotonin antagonists**
- If refractory to monotherapy, combine antiemetics with different mechanisms

TYPES OF NAUSEA AND VOMITING



Motion Sickness



Post-Operative Nausea/Vomiting (PONV)



N/V of Pregnancy (NVP)



Chemotherapy-Induced Nausea/Vomiting (CINV)

MOTION SICKNESS

- Normal physiologic response to unusual perception of motion (visual-vestibular mismatch)
- Occurs due to **abrupt changes in motion**
 - Car sickness
 - Air sickness
 - Sea sickness
- Involves the **inner ear and vestibular system**
- **Acetylcholine and histamine** thought to be the most responsible neurotransmitters signaling the medulla oblongata

MOTION SICKNESS

Prevention > Treatment

- Adequate hydration, avoidance of smells/foods that induce nausea, eating frequent small meals instead of large meals, relaxation therapy, pain treatment
- May be less severe when driving, if possible, or sitting in the front seat with a clear forward view

Treatment → targets histamine and acetylcholine

- Take medication 30-60 minutes prior to motion exposure
 - Dimenhydrinate (Gravol®) if ≥ 2 y/o or diphenhydramine (Benadryl®) if ≥ 6 y/o
 - Are most effective for short durations
 - Onset of 15-30 min and lasts 4-6 hours
 - ADRs (common for both): sedation, anticholinergic effect (e.g. dry mouth, constipation, urinary retention, etc.)
 - ADR (dimenhydrinate/Gravol®): Paradoxical excitation in children.
 - Promethazine → longer duration of action than dimenhydrinate

**Ginger & acupressure wristbands (SeaBands®) have not consistently demonstrated benefit, although widely promoted for antinausea*

POSTOPERATIVE NAUSEA AND VOMITING (PONV)

- Occurs within 24-hours after surgery
- Includes all known pathways
- 4 Key Patient Risk Factors

Incidence of PONV increases incrementally with each additional risk factor (10%-80%)

1. Female sex
2. Non-smoker
3. History of motion sickness/PONV
4. Opioid use



- Other risk factors include:
 - Type of anesthetic (nitrous oxide, volatile anesthetics)
 - Duration of surgery
 - Dehydration
- Reducing risk factors is most effective management strategy
- Preventative therapy with anti-emetics in moderate-high risk patients

POSTOPERATIVE NAUSEA AND VOMITING (PONV)

Pharmacological Options:

Combination therapy is more effective due to numerous neurotransmitters involved

- Ondansetron
- Haloperidol
- Dexamethasone
- Metoclopramide
- Dimenhydrinate

Non-Pharmacological Options:

- Stimulate P6 acupuncture point
- Remain hydrated
- Avoid noxious odours

NAUSEA AND VOMITING OF PREGNANCY (NVP)

- Referred to as “morning sickness”
 - Can occur at any time and may be constant throughout
- Pathophysiology is unclear
- Occurs in approximately 50-80% of pregnancies
- Normal to occur during **first trimester** of pregnancy – generally subsides by 16 weeks
- Severe, persistent NVP is called hyperemesis gravidarum (HG)
 - Occurs in approximately 2% of pregnant women
 - Can continue throughout the entire pregnancy
 - Leads to weight loss, electrolyte imbalances, dehydration, nutritional deficiency
 - **Must refer**

NAUSEA AND VOMITING OF PREGNANCY (NVP)

Pharm Options*:

- 1st line for mild NVP: **Diclectin®** (pyridoxine/doxylamine) or **pyridoxine** (Vitamin B6) alone
 - Can add **dimenhydrinate** if no improvement
- If no improvement/ symptoms persist (moderate-severe NVP or persistent-mild or hyperemesis gravidarum):
 - Without dehydration – Add: **metoclopramide** (for up to 5 days) or **chlorpromazine** or **prochlorperazine** or **promethazine**
 - With dehydration – Rehydration (IV fluid replacement) +/- therapies above depending on symptom control
- For refractory cases: can add **ondansetron** as adjunctive when other therapies are not effective
- Reserve **methylprednisolone** IV for resistant cases after the first trimester

* Can be used at any stage: pyridoxine, ginger, P6 acupressure, non-pharmacological interventions

Non-Pharm Options:

- **Avoid triggers:** Spicy food, foods with high fat content, fried food, strong food odours
- **Eat small and bland meals**
- May discontinue iron-containing prenatal vitamins during the first trimester and substitute with folic acid or vitamins low in iron
- Prevent dehydration and electrolyte imbalance
 - Fluid intake, rehydration solutions if severe and/or accompanied by diarrhea)
- Mindfulness-based CBT

PHARMACOLOGICAL MEASURES- OVERVIEW

NON-PRESCRIPTION THERAPY

- Dimenhydrinate (Gravol®)
- Diphenhydramine (Benadryl®)
- Ginger
- Pyridoxine (Vitamin B6)

PRESCRIPTION THERAPY

- Doxylamine succinate / pyridoxine (Diclectin®)
- Domperidone
- Metoclopramide
- Haloperidol
- Dopamine Receptor Antagonists (phenothiazines)
- 5-HT₃ Receptor Antagonists (5-HT₃ RA) (Ondansetron)

PHARMACOLOGICAL MEASURES- NON-PRESCRIPTION THERAPY

Dimenhydrinate (Gravol®)

- First-line therapy for motion sickness (useful for short duration of exposure)
- Used in persistent or moderate-severe NVP (in addition to Ginger or pyridoxine)

Diphenhydramine (Benadryl®)

- First-line therapy for motion sickness (useful for short duration of exposure)
- May be used in persistent or moderate-severe NVP (in addition to Ginger or pyridoxine)
 - Dimenhydrinate > diphenhydramine in NVP

Tips:

- **Onset: 30 mins**
- Effective in all types of N/V except chemotherapy-induced
- **Side effects:** dry mouth, blurred vision, constipation, sedation, urinary retention, increased intraocular pressure, paradoxical excitation in children

PHARMACOLOGICAL MEASURES- NON-PRESCRIPTION THERAPY

- **Ginger** (Gravol Natural Source®)
 - First line in mild NVP (commonly used as 250 mg QID)
 - Used for motion sickness, PONV
 - **Side effects:** GI discomfort, diarrhea, dyspepsia, pepper-like irritant effect in mouth/throat
 - May inhibit in vitro CYP2C19; clinical significance unknown
- **Pyridoxine** (Vitamin B6)
 - First line in mild NVP
 - May be used alone or combined with doxylamine (Diclectin)
 - Used to treat nausea and vomiting in pregnancy (NVP)
 - **Onset:** 1 – 2h
 - **Side effects:** GI pain, headache, loss of appetite, photosensitivity, somnolence

PHARMACOLOGICAL MEASURES- PRESCRIPTION THERAPY

- Doxylamine succinate / pyridoxine (Diclectin®)
 - Antihistamine in combination with vitamin B6
 - 1st line therapy for NVP
 - Take at regular intervals (up to QID, delayed release tablet)
 - Side effects (antihistamine): + + + sedation, diarrhea, disorientation, headache, epigastric pain, palpitations, urinary retention
 - Drug interactions: MAOIs, antimuscarinics, CNS depressants

PHARMACOLOGICAL MEASURES- PRESCRIPTION THERAPY

- Domperidone (dopamine antagonist)
 - Do not use in pregnancy!
 - Treatment for functional dyspepsia, gastroparesis
 - Does NOT cross BBB
 - Concern for QT prolongation, serious ventricular arrhythmias, hyperprolactinemia
- Metoclopramide (dopamine antagonist)
 - Treatment for drug-induced nausea, migraine-induced nausea, gastroparesis, NVP (persistent), CINV
 - Crosses BBB
 - Larger concern for EPS and tardive dyskinesia
 - Interactions with dopaminergic medications/disease states (i.e. Parkinson's disease)

Tips:

- Side Effects: diarrhea, abdominal cramps, distension, headache, hyperprolactinemia
- Caution sedation with CNS depressants (e.g. alcohol)

PHARMACOLOGICAL MEASURES- PRESCRIPTION THERAPY

Haloperidol (dopamine antagonist)

- Low doses
- Treatment of PONV and refractory CINV (less effective than metoclopramide for refractory CINV)

Tips:

- **Side effects:** QT prolongation, sedation, EPS
- Caution when used with other drugs that cause QT prolongation
- Caution in drugs/disease states that have dopaminergic involvement (i.e. Parkinson's, other antipsychotics or prokinetics)
- Highest likelihood of causing EPS
- May have mild anticholinergic side effects

PHARMACOLOGICAL MEASURES- PRESCRIPTION THERAPY

Dopamine Receptor Antagonists (Phenothiazines)

- Chlorpromazine
 - Persistent NVP
- Perphenazine
 - Treatment of PONV
- Prochlorperazine
 - Treatment of PONV, CINV, drug-induced nausea, persistent NVP
- Promethazine
 - Motion sickness, PONV, gastroenteritis, moderate-severe NVP and persistent NVP

Tips:

- **Side Effects:** anticholinergic effects, sedation, hypotension when administered IM or IV
- EPS/QT prolongation rare but possible, especially with higher doses

PHARMACOLOGICAL MEASURES- PRESCRIPTION THERAPY

5-HT₃ Receptor Antagonists (5-HT₃ RA)

- Granisetron
- Ondansetron
 - Decreases analgesic effects of tramadol
 - CYP3A4 inducers decreases [ondansetron]
- Palonosetron

Tips:

- Treatment for PONV and acute CINV, alternative for persistent, moderate-severe NVP
- **Side effects:** constipation, headache, diarrhea, QT prolongation, bradycardia, dizziness
- Risk for serotonin syndrome when used in combination with other serotonergic medications

TAKEAWAY

- Nausea/vomiting (N/V) is a symptom, not a diagnosis and can have various underlying causes
- **Red flags** include significant dehydration, bleeding, altered mental status, head trauma, thunderclap headache, fever, stiff neck. These red flags **warrant referral**
- **Motion sickness**: related to acetylcholine and histamine signaling
 - Treat with **diphenhydramine, dimenhydrinate, or promethazine** 30-60 minutes prior to motion
- **Nausea and vomiting of pregnancy (NVP)**: also known as morning sickness
 - Mild → **Diclectin®** or **pyridoxine**. Can add dimenhydrinate if no improvement
 - If no symptom improvement/persistent symptoms → **metoclopramide** or **chlorpromazine** or **prochlorperazine** or **promethazine**. If dehydration, rehydrate with IV fluids
 - Non-pharmacological interventions: avoid triggers, small bland meals, adequate fluid intake
- **Postoperative Nausea and Vomiting (PONV)** occurs within **24 hours** of surgery
 - Main risk factors: female sex, non-smoker, history of motion sickness/PONV, opioid use
 - Treatment: **Manage risk factors** (e.g. avoid noxious odors, hydration). If pharmacological management is considered, **combination therapy is more effective**
- **BE MINDFUL OF DRUG INTERACTIONS!**
 - Most antiemetics cause some degree of sedation and drowsiness. Use caution when prescribed alongside CNS depressants (e.g. alcohol, opioids, benzodiazepines)

CASE 1

AS is a 31-year-old female who presents to your community pharmacy with complaints of nausea. AS is in her first trimester of pregnancy. The nausea occurs consistently throughout the day, although it is worse in the morning, and leaves her unable to eat. She has tried acupressure and antacids but has not achieved any relief. She is open to trying medications and would like your advice on how to best manage her nausea. AS has a past medical history of dysmenorrhea. She takes PregVit® (multivitamin) once daily and an antacid (calcium carbonate) PRN. She has a history of motion sickness, especially when travelling on boats. She does not smoke or drink alcohol and she has a documented allergy to penicillin, having experienced a rash as a child.



CASE 1 - QUESTIONS

1. Which of the following should NOT be recommended to AS?
 - a) Take supplements to increase her intake of iron
 - b) Eat frequent, low-fat meals
 - c) Drink small amounts of fluid throughout the day
 - d) Avoid perfumes and other sources of strong odours

2. Which of the following anti-emetic options is the BEST recommendation for AS?
 - a) Promethazine 12.5 mg Q6H
 - b) Metoclopramide 10 mg PO Q4H PRN
 - c) Pyridoxine 25 mg PO TID PRN
 - d) Diphenhydramine 25 mg Q4H PRN

CASE 1 - QUESTIONS

3. Which of the following are signs of dehydration that AS should monitor for?
 - a) Reduced skin turgor, decreased urination, urticaria
 - b) Sunken eyes, fever, dry mouth and skin
 - c) Increased thirst, reduced skin turgor, fatigue
 - d) General malaise, drowsiness, increased thirst

4. Which are important counselling points if AS were to initiate doxylamine succinate/ pyridoxine?
 - a) Take 1 tablet on a PRN basis at the first sign of nausea
 - b) Patients should not drive or operate machinery after taking this medication
 - c) It should be used concurrently with scopolamine to maximize its anti-emetic effects
 - d) Dimenhydrinate is first-line and should be tried first before starting doxylamine/ pyridoxine

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CHANGE LOG

August 2025

- Formatting and grammatical changes made throughout
- Slide 14: Removed duplicated information
- Slide 20: Added note on discontinuing iron-containing prenatal vitamins
- Slide 19: Added a note about referral

November 2025

- Minor formatting and grammatical changes throughout
- Slide 29: Updated takeaway

March 2026

- Slide 2: Updated competencies
- Slide 8: Added antidepressants
- Slide 13: Mentioned poor intake as reason for ORS
- Slide 15: Visual Vestibular Mismatch in motion sickness. Discussed position in car in relation to motion sickness
- Slide 18: Minor updates
- Slide 20: Updated recommendations
- Slide 24: Frequency for Diclectin
- Slide 29: Updated takeaway
- Slide 31: Updated answer choices of question 2